

A Geometrical Theory of Communication: Measurement, Meaning and Representation

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Abstract

What makes an alternative identifiable before we count it? Shannon's information measure begins with distinguishable possibilities and assigns an amount to the uncertainty of their selection. I trace that measurement back to the comparison it uses: a held reference, distinct readings, and a unit that remains the same when repeated. The geometric bit brings these relationships together through the circle, where perpendicularity, symmetry, and closure give identity an operational form. Euler's formula expresses their composition. The argument places this geometric identity at the foundation of counting and communication, then follows three questions: what exists, what do we measure/observe, and how do we convey it? Physical interaction, measurement, division algebras, Hopf projections, and language enter as parts of the account of how an identity becomes available, survives a change of expression, and is recovered by another observer.

Introduction

What is information? What does that mean?

Ask anyone and you get an answer, quickly: news, data, knowledge, bits, something I did not know before. Everyone reaches for it, everyone reaches somewhere a bit different but the word keeps working, as it has been doing so for a long time: *informare*, to give form, to shape, to inform someone is to give a shape to what they know. Shannon gave us a precise answer in *A Mathematical Theory of Communication*: a measure of uncertainty, defined through entropy, that set the quantitative foundation of information theory and modern communication (Shannon, 1948). But uncertainty about what? Does that answer our question about what information is? Bear with me as we unpack this question thoroughly.

Asking what information is already assumes that the question has an object—that the word points to something, whether we call it data or knowledge, it's an object that exists in some sense, even if abstract. Let us say it does, and let us say that I, the author, have seen it and measured it, and so I know what information is. How do I transfer that knowledge to you, the reader? How do I give you information about information? And how do you know that knowledge is true?

Weaver already wrote down the three levels of communication problems (Shannon and Weaver, 1949, pp. 4–5). Word for word:

Level A — Transmission. How accurately can the symbols of communication be transmitted? (The technical problem)

Level B — Meaning. “How precisely do the transmitted symbols convey the desired meaning?” (The semantics problem)

Level C — Effectiveness. “How effectively does the received meaning affect conduct in the desired way?” (The effectiveness problem)

The original mathematical theory solved Level A, and Shannon drew that boundary himself: the semantic aspects of communication “are irrelevant to the engineering problem” (Shannon, 1948, p. 379). He was right about the engineering: the mathematics works, and a great deal of current technology runs on it. Information came away from that work with a number attached, our computational bit, while most of our questions about what information actually is remain open. The level B problem is exactly what this paper is going after.

Let’s say I point to a lighter on the desk between us: we both see it, and we both know that it is there. What does that mean? The word *lighter* points to an object in the world, and our two perspectives on that object still point the same word at the same thing. I could call it an igniter and you would receive the same information through a different symbol, the word carries a meaning we can already both reach.

The physical object, our perspective on it, and the symbol we use to represent it. three questions separate the objects here, and together they answer what information is. What exists as a physical entity? How do we observe, measure or know it? How do we represent and transfer that knowledge?

This is a unavoidably an interdisciplinary subject: for what counts as existing, we need physics; to know that it holds independently of my beliefs, we need epistemology; to convey it, we need linguistics, mathematics and Shannon’s original paper—because as soon as I know it and set out to tell you, we are already inside Level B, how do I give you information about information through my own set, and how can you know it’s true?

The first part of this paper works through an object where the separation between reality, measurement and notation stays visible: a circle. Closure over rotation, written by Euler as a mathematical identity, supplies the preserved relationship we need to answer the question of information in a precise and robust way. Shannon’s definition of the bit aimed at the same target, and his description returns here as a partial case, pairing entropy with geometric closure. That yields a definition of a *geometric bit*: a unit of comparison that retains a distinction together with the reference needed to interpret it.

If I say A, you must understand A for that meaning to have been communicated. The agreement between *igniter* and *lighter* already depends on this: the expressions differ, while what they mean in that encounter is the same. Language allows variation in how much is understood, but communication requires this shared ground; scientific descriptions depend on agreement precise enough to identify the same subject and follow the same claim; and for mathematical objects, their defining relationships make that requirement exact—an identity.

Defining the geometric bit gives the principle of identity, $A = A$, an operational form: the identity required for meaning to pass from one person to another is built into the unit itself. This connects the two parts of the paper: establishing what information is establishes the identity to be conveyed, and the theory of its transmission asks how that identity is preserved between sender and receiver.

The second part asks how to send it, now with the unit in hand: what must a description retain for a reader to recover the comparison it was about, how do several comparisons combine, and how much of one survives noise and limited precision?

A note on interdisciplinarity

Where should an account of information begin if its meaning depends on all three questions? A record has a physical realization, as Landauer emphasizes (Landauer, 1996), and obtaining a reading requires an interaction with what is being observed. Quantum information makes this connection especially clear by relating what can be distinguished to the states and

measurements a physical system allows (Nielsen and Chuang, 2010). Physics enters the inquiry wherever a claim about information depends on how a distinction is made in the world.

Expressing that distinction mathematically makes its relationships precise enough to calculate and compare. Computing and machine learning ask how representations can be used and learned; psychology asks how people learn to recognize what a sign means; philosophy and epistemology ask what grounds that recognition as knowledge. These concerns meet in Level B, where checking whether meaning was conveyed requires both an account of what was meant and a way of recognizing its recovery.

The same problem applies to the paper itself: its symbols are meant to convey relationships that a reader can understand and check. Information theory has a metascientific role in this sense, since every science records observations and communicates what is known through descriptions. The interdisciplinary approach follows those connections, from what exists, through how it becomes known, to how that knowledge becomes available to someone else.

1 From Shannon's bit to the geometric bit

How did Shannon and Weaver make the transmission problem measurable? They set meaning aside on purpose: a message full of meaning and a string of nonsense can have the same information measure, because the task is to recover at one end the message selected at the other. The uncertainty concerns which of the possible messages was chosen, so the measure belongs to the whole choice situation (Shannon and Weaver, 1949, pp. 8–9).

1.1 Shannon's bit: what the choice already assumes

Weaver's example makes this easy to see. The two possible messages are the whole King James Bible and the single word "Yes." One is a book and the other is a word, yet either can be selected by the same small signal: we agree beforehand that 0 selects the Bible and 1 selects "Yes," or that a closed relay selects one and an open relay the other. I already have the messages and their labels; when your signal arrives, I learn which one you chose.

With the alternatives equally likely, we assign this choice the value one and call its unit a bit, John Tukey's contraction of *binary digit*. Weaver calls the assignment a convention: the unit measures the choice between the two messages, whatever their length or meaning (Shannon and Weaver, 1949, p. 9).

What happens when we have more choices? Two independent binary choices give four possible pairs, and three give eight possible triples. The possibilities multiply while the information adds, which is the counting rule expressed by the logarithm. Unequal probabilities change how much we learn: an almost certain answer is less surprising than a rare one, and entropy averages over these possibilities (Shannon, 1948).

Formal 1.1: The choice, its unit, and entropy

For a finite set of N equally likely messages, the information is $I(N) = \log_2 N$, with $I(2) = 1$ fixing the unit. Independent choices multiply the number of possibilities, so

$$I(NM) = I(N) + I(M).$$

The logarithmic form follows from additivity and regularity. For a continuous extension J to positive arguments satisfying $J(xy) = J(x) + J(y)$, the function $g(t) = J(e^t)$ is continuous and additive, hence $g(t) = ct$. Thus $J(x) = c \ln x$, and $J(2) = 1$ gives $c = 1/\ln 2$.

For a message m of probability $p(m) > 0$, its self-information is $-\log_2 p(m)$. The source entropy is its expectation,

$$H(M) = - \sum_m p(m) \log_2 p(m), \quad 0 \log_2 0 := 0.$$

A binary source has $0 \leq H(M) \leq 1$, with equality at the upper bound precisely for equal probabilities. Two available states and one bit of entropy coincide in that case (Shannon, 1948).

How does the bit look geometrically? Put 0 and 1 on a line, holding 0 as the reference and placing 1 one unit away. Reading the mark means comparing its position with that origin and unit. If we move along the line, the reading changes; if we move along the perpendicular through 1, it stays at 1, just as a ruler's tick carries a reading across the page.

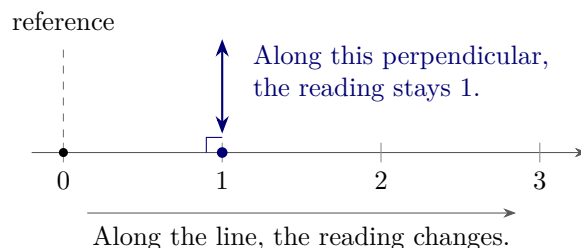


Figure 1: A reading and its reference. Movement along the horizontal line changes the reading; movement along the perpendicular through 1 preserves it. The interval from 0 to 1 fixes the unit.

Extend the line to 2 and 3, and the same comparison repeats: the interval from 1 to 2 must agree with the interval from 0 to 1. Translation carries one interval onto the next while preserving its length. In this physical account of counting, perpendicularity keeps the coordinate readings separate and symmetry keeps the repeated step the same, whether or not we draw every tick.

Formal 1.2: What the perpendicular preserves

Fix a Euclidean plane, an origin o , a unit direction u , and a length unit $\ell > 0$. The reading of a point p along that direction is

$$x(p) = \frac{\langle p - o, u \rangle}{\ell}.$$

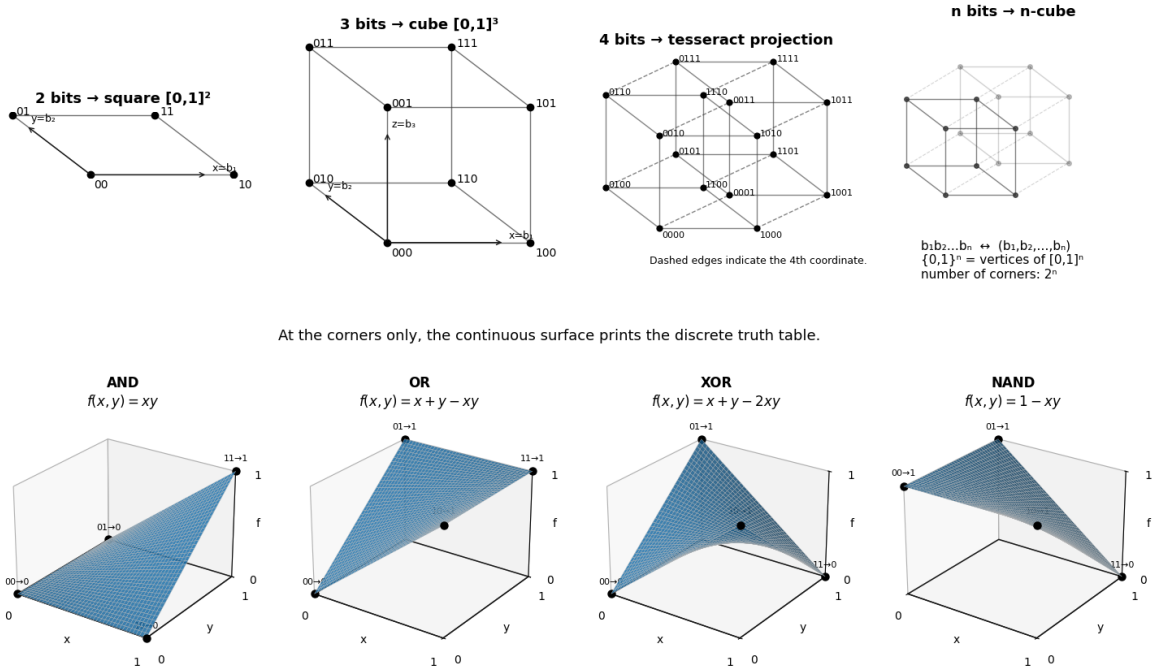
A displacement v preserves the reading exactly when

$$x(p + v) - x(p) = \frac{\langle v, u \rangle}{\ell} = 0, \quad \text{that is, } v \perp u.$$

Thus each fixed reading is a line perpendicular to u . The points $p_k = o + k\ell u$ have readings k , and translation by ℓu sends p_k to p_{k+1} while preserving distances. These are the projection and equal-step properties used by the ruler construction.

What happens when we put two bits together? Each must be able to change while the other stays fixed, giving the four readings 00, 01, 10, and 11 at the corners of a square. A third bit adds another perpendicular coordinate and gives the eight corners of a cube; each further bit adds another direction in which the record can vary. An n -bit string is one corner of this cube, read from the common starting corner $00 \dots 0$ (O'Donnell, 2014, Chapter 1).

Bit strings as cube corners; Boolean gates as exact multilinear extensions



Convention used throughout: bit string $b_1b_2\dots b_n$ maps directly to coordinate point $(b_1, b_2, \dots, b_n)$. Other bit-order conventions are equivalent up to axis permutation.

Figure 2: Binary states and their operations read geometrically. The upper row follows independently variable bits into squares, cubes, and higher-dimensional cubes. Below, each surface gives a gate's output at its input corners. The surfaces are the corresponding multilinear extensions of the truth tables.

Computers already use this geometry in their operations. On the square in Figure 2,

AND returns one only at 11, just as multiplication xy does. XOR returns one at 01 and 10, where the inputs differ, and zero where they agree. Each gate reads a corner and supplies an output that another gate can use, allowing these small comparisons to combine into a computation.

Formal 1.3: Gates as functions on the cube

The binary state space is $B_n = \{0, 1\}^n \subset \mathbb{R}^n$, with 2^n vertices and one coordinate per bit. A Boolean gate $f : B_n \rightarrow \{0, 1\}$ has the unique multilinear extension

$$F(x) = \sum_{a \in B_n} f(a) \prod_{j: a_j=1} x_j \prod_{j: a_j=0} (1 - x_j).$$

At a vertex b , the product indexed by $a = b$ is one and all others vanish, so $F(b) = f(b)$. These products form a basis for the multilinear polynomials, establishing uniqueness within that class. For two inputs,

$$F_{\text{AND}}(x, y) = xy, \quad F_{\text{XOR}}(x, y) = x + y - 2xy.$$

The surfaces in the figure are these extensions; the gate's binary outputs are their values at the corners (O'Donnell, 2014, Section 1.2).

The cube also tells us how far apart two messages are. A flipped bit moves a record along one edge, so the shortest route between two corners counts the positions where their strings differ. That count is Hamming distance, and zero means the same string. To correct errors, choose valid records far enough apart that a few wrong bits still leave the received corner closest to the one that was sent (Hamming, 1950).

Formal 1.4: Distance and recovery

For $x, y \in B_n$,

$$d_H(x, y) = \sum_{j=1}^n \mathbf{1}_{x_j \neq y_j} = \sum_{j=1}^n (x_j - y_j)^2 = \|x - y\|_2^2.$$

Each summand is zero or one. This is also the shortest edge-path length on the cube: each differing coordinate must be flipped at least once, and flipping each once reaches the destination. For a code $C \subseteq B_n$ with at least two codewords, let $d_{\min} = \min_{c \neq c' \in C} d_H(c, c')$. If $d_{\min} \geq 2t + 1$, nearest-codeword decoding corrects any pattern of at most t bit errors. A word within t of two codewords would, by the triangle inequality, put them at distance at most $2t$, contradicting the separation (Hamming, 1950).

Some computers even used the cube to connect their hardware. The Connection Machine CM-1 linked 4,096 routing nodes in a 12-dimensional Boolean cube, with neighbouring addresses differing in one bit. Moving a message between nodes followed the same coordinate changes used to compare their addresses (Hillis, 1985, Section 4.3).

What, then, has the bit given us? A difference recorded against a fixed reference, a way to combine such records, and operations for computing with them and recovering them after noise. The agreement about what each record selects remains in place throughout. If you send 0 for the Bible while I read 0 as “Yes,” the string arrives correctly and I still recover the wrong message.

Formal 1.5: The shared agreement

For the agreed repertoire $\mathcal{M} = \{\text{Bible}, \text{Yes}\}$, let the sender's encoding be $E_S : \mathcal{M} \rightarrow \{0, 1\}$, with $E_S(\text{Bible}) = 0$ and $E_S(\text{Yes}) = 1$. Let $D_R : \{0, 1\} \rightarrow \mathcal{M}$ be the receiver's interpretation. If $b = E_S(m)$ is sent and $\hat{b}$ is recovered, the two checks are

$$\hat{b} = b, \quad D_R(\hat{b}) = m.$$

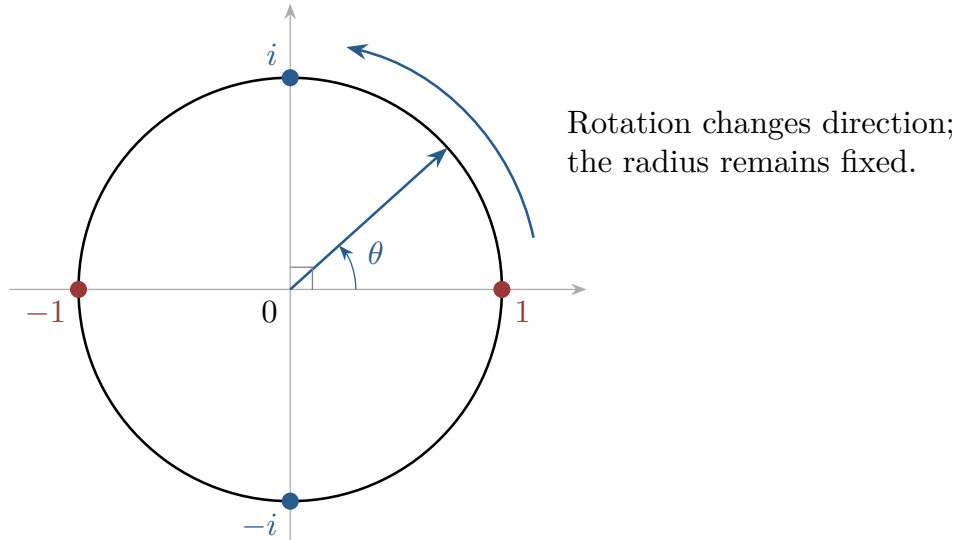
With exact symbol recovery, agreement for every message requires $D_R \circ E_S = \text{id}_{\mathcal{M}}$. This equation specifies recovery within the agreed repertoire; the assignment of labels to its members is part of the setup. Reversing D_R preserves symbol recovery while making the second check fail.

Every transmitted record therefore leaves us with two questions: did the receiver recognize the state that was sent, and did that state select the same thing for both of us? The first is Shannon's technical problem; the second is Weaver's Level B (Shannon and Weaver, 1949, pp. 4–5). How can that agreement become something we perform and check, instead of something we assume when assigning the labels?

1.2 The geometric bit: identity through an operation

Draw a circle and mark a starting point. Its boundary separates inside from outside, and following it brings us back to the mark after a full turn. We can watch the position change throughout the movement and compare the result with the reference from which we began.

Imagine the circle as a pizza to be divided into four equal pieces. Do we need a compass to recognize where the cuts should go? A diameter and its perpendicular give four right-angle sectors, a relationship we can see before drawing the lines. Their opposite ends give four poles, and turning the arrangement keeps the cuts at right angles and the poles opposite.



Perpendicular directions distinguish components.

Opposite poles share the same radius.

Four quarter-turns restore the marked direction.

Figure 3: One construction joins distinction and preservation. The perpendicular axes separate component directions; each axis has opposite poles; rotation changes their orientation together while preserving their lengths and angles. A mark distinguishes the positions through which the circle can be traversed.

Take a reading, turn the arrangement, and compare it with the first. The marks have moved, while the radius and right angles remain. We can check whether the drawing has rotated or been distorted by measuring these lengths and angles. A quarter-turn takes one pole to its perpendicular, a half-turn takes it to the opposite pole, and a full turn brings it back to the starting point.

What happens if you begin from a different direction? Put your reference at the top of the circle while I put mine on the right, and let us both make a quarter-turn in the same direction. Our marks finish at different places, but each has turned through the same angle relative to its own reference. Rotating a reference and its reading together adds the same angle to both; their difference stays fixed. We can compare our results because the reference travels with the reading, and the turn carries its own direction in the order its positions are reached.

Now change the size of your circle. Its circumference and diameter grow together, keeping the ratio π , while the right angles and the fraction of a turn remain unchanged. We can use different radii and different starting directions and still check the same quarter-turn. The shared result is available through the comparison we perform, rather than through agreement on a particular drawing.

To follow this movement continuously, hold the radius fixed and let its direction turn. Successive angles add, giving the same result as the combined turn performed at once. The exponential expresses this continuous composition; for a pointer turning at a fixed rate, elapsed time determines how much angle has accumulated. A complete traversal supplies an event we can record and repeat.

Where is the point now, and how many turns has it made? These are different readings of the same movement. After each full turn the point is back at its reference, while the retained count increases by one. Each repetition uses the same completion criterion, just as each interval on the numbered line uses the same length. Addition collects these repetitions, and multiplication collects equal groups of them.

Let 1 mark the initial unit radius and multiplication by i perform a positive quarter-turn. Two applications reach the opposite direction, -1 , giving $i^2 = -1$; the third reaches $-i$, and the fourth returns to 1. Euler's formula describes the intermediate angles, with the half-turn and full-turn values written as

$$e^{i\pi} + 1 = 0, \quad e^{2\pi i} = 1.$$

The first records opposition through cancellation; the second records the completed return. Starting with any marked radius A , a full turn gives $e^{2\pi i} A = A$: we have an operation through which the equality can be checked.

I call this repeatable comparison the *geometric bit*. Shannon's transmission problem takes the identities of the alternatives and their agreed interpretation as given, then checks whether the selected symbols are recovered. The geometric bit makes $A = A$ operational by carrying the reference, the transformation, and the comparison through which the result is recognized. This is the account of identity developed here for Level B: another person can follow the expression and reproduce the relationship it describes.

For example, you can draw the two perpendicular diameters while I follow four quarter-turns. We can compare where the poles lie, the angles between them, and the return to the starting point. The drawing and the formula direct us to the same checks, allowing us to establish what we understood through what we can reproduce.

To take a binary reading from the circle, record which side of a diameter contains the mark, with a convention for points on the diameter. Using two perpendicular diameters gives four quadrants; we can retain their labels while setting aside the exact angles. Once these

alternatives and their probabilities are specified, Shannon's measure applies to the choice among them.

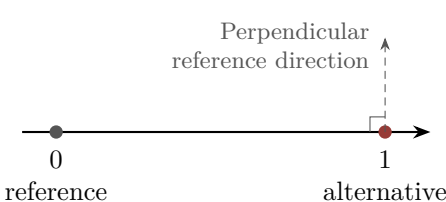
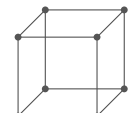
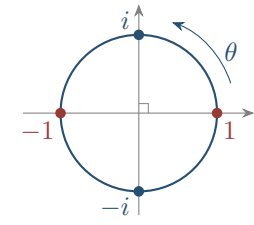
What follows when we compose turns about different spatial axes? Their order can change the result, so we must keep track of which turn came first. Quaternions describe this composition while preserving lengths. The normed division algebras extend the construction from complex numbers through quaternions to octonions, retaining the earlier algebra inside the next and preserving the unit norm under multiplication (Baez, 2002).

The associated Hopf maps describe what a projection leaves visible: several states give the same reading, and the alternatives left unresolved form its fibre. In this family, the total space at one level reappears as the fibre type at the next. Section 2 follows this construction beyond the circle on the page into spatial orientation and its changing views.

Figure 4 brings the two descriptions together. We began with a signal whose alternatives and interpretation were already available, then followed the counting back to a unit we could repeat and check. To ground that comparison and carry it between observers, we return to the questions from the introduction: what exists, what do we measure/observe, and how do we convey it?

Two bits. Two geometries.

From the binary record to the relationships its comparison uses

Shannon's bit A 0–1 line  Which alternative? What it encodes • Discreteness: 0 or 1. • Difference: reference and alternative. • One binary coordinate. Geometric interpretation A distinction along a line. The reading fixes a reference and records which of the two states is selected. Composition Bits $\rightarrow$ registers Gates $\rightarrow$ computations Resulting geometry Hypercubes Square $\rightarrow$ cube $\rightarrow \dots$ n bits: 2^n vertices.  What it gives us • Classical information measures. • Digital computation. • Discrete state machines.	The geometric bit A circle — Euler's relationship  What makes it the same through change? What it encodes • Discreteness: inside and outside. • Difference: perpendicular directions. • Symmetry: opposite poles. • Space: π relates path and diameter. • Time: $e^{i\omega t}$ accumulates rotation. • Identity: the relationship is preserved. Geometric interpretation A comparison around a circle. Radius and right angles are preserved. Opposite poles remain opposite. Composition Circular rotations $\rightarrow$ spatial rotations Algebraic extension: $\mathbb{R} \subset \mathbb{C} \subset \mathbb{H} \subset \mathbb{O}$ Resulting geometry Spheres and Hopf projections S^1, S^3, S^7, S^{15} $S^3 \rightarrow S^2, S^7 \rightarrow S^4, \dots$ Spheres drawn schematically.  What it gives us • A geometric account of identity. • Comparison through transformation. • Identity, measurement, and meaning.
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The binary record and its geometric ground.

Figure 4: From the binary distinction to the geometric comparison. The aligned panels compare the binary record with the relationships that this paper places at the ground of its counting and comparison. The right column develops this account through the geometric bit and its composition.

2 What exists?

The lighter on the desk reflects light, resists a hand, exchanges heat, and can produce a flame. Those interactions are how we encounter it. What would it mean to say that something existed physically if it could enter no interaction at all?

The starting premise of this account is that to exist physically is to be available to interaction. Someone need not be observing at the time. The cup remains on the desk when we turn away; its relationships with the desk, the air, and the surrounding light continue.

We are part of this picture too. The observer, the instrument, the inscription, and the act of thinking all occur in spacetime. Imagining a blue lighter is an activity of a physical observer, whether or not a blue lighter is on the desk. The occurrence of the thought and the presence of its object are separate questions (Russell, 1905). An abstraction becomes available through a physical activity, and a record has a physical realization (Landauer, 1996).

How does that activity connect an object with a description? The inscription “10” could be a length, a mass, or a reading from an instrument with the wrong calibration. The number alone cannot tell us. We need the thing, a reference, and the comparison between them. The interaction is the third term that makes the reading interpretable. Peirce’s account of object, sign, and interpretant gives a precedent for this triadic structure (Peirce, 1932, §§2.228, 2.274).

2.1 The cup and its changing picture

Look straight down at a round cup. Its rim appears circular. Tilt it and the rim appears elliptical; look nearly along its plane and the opening becomes a narrow shape. Has the cup changed? Its spatial relationship remains while the view changes.

A single picture leaves the viewing direction unresolved. A different shape seen from another angle can give the same outline. Moving the cup, or moving around it, supplies further views. Their agreement lets us recognize one object through changing appearances.

This is the $3 + 1$ encounter in ordinary terms: an object in three spatial dimensions, observed as it changes through time. The circle on the page is one view of a relationship realized there. The observer’s direction and the order of the observations matter when we reconstruct it.

There is a related distinction in dynamics. Poincaré–Bendixson shows how strongly a smooth autonomous motion is constrained when its state has only two dimensions. If an orbit stays in a compact part of the domain and its limiting set contains no equilibrium, that limiting set is a periodic orbit (Sideris, 2013, Theorem 8.8). The original motion may approach the cycle without exactly repeating. Three state dimensions allow bounded chaotic behaviour, as in the Lorenz system (Tucker, 2002). An enduring comparison must therefore remain usable even when the process being compared does not return.

The three roles in a comparison and the dimensions of a dynamical state answer different questions. They meet here in the physical problem: how can a reference remain available while the thing being observed changes?

2.2 Composing the turns

Turn the cup about one axis and then another. Reverse the order and its final orientation can change. A description that forgets the order cannot tell us which orientation was reached.

Quaternion multiplication retains this order. Unit quaternions describe spatial rotations whose action preserves lengths and right angles. Each individual turn still has the circular form: a chosen axis and an angle around it. The product tells us how turns about different axes combine (Baez, 2002).

The scalar and three imaginary components describe this rotation; following them through time describes a changing orientation. Their use in the physical encounter gives the planar circle a spatial setting. A relativistic account additionally specifies the spacetime metric with which events are compared.

The division-algebra construction extends the relationships by forming pairs and carrying multiplication and conjugation into the new pair. This is Cayley–Dickson doubling. It gives the real numbers, complex numbers, quaternions, and octonions as the four real normed division algebras. Their defining norm rule is simple: the norm of a product is the product of the norms. Unit elements consequently compose to unit elements (Baez, 2002).

Order matters for quaternions. For octonions, grouping matters too: changing the parentheses can change a product. The rules say which relationships a composed description must retain. Inside each larger algebra, a chosen complex plane still carries Euler’s circular relationship.

2.3 What the Hopf projection leaves visible

The Hopf maps organize a comparison and what it leaves unresolved. A point in the base is the returned description. The fibre contains the states that give that same description. In the complex case, for example, normalized pairs differing only by a common phase give the same result; the unresolved phases form a circle.

The family has the following form (Baez, 2002):

Algebra	Fibre	Total space	Base
$\mathbb{R}$	S^0	S^1	S^1
$\mathbb{C}$	S^1	S^3	S^2
$\mathbb{H}$	S^3	S^7	S^4
$\mathbb{O}$	S^7	S^{15}	S^8

Here S^n means the n -dimensional sphere. Notice what happens between rows: the total space of one comparison becomes the fibre type of the next. A space that held the whole preceding comparison reappears among the alternatives of a richer one.

The recurrence is exact. For a normalized pair from any of the four algebras, the map can be written

$$h(a, b) = (|a|^2 - |b|^2, 2a\bar{b}), \quad |a|^2 + |b|^2 = 1.$$

The norm rule places its output on the base sphere. Embedding a lower algebra into a higher one preserves multiplication, conjugation, and norm, so the higher map reproduces the lower map on those embedded states. At a distinguished fibre, the previous total space is identified directly through the doubling coordinates.

This recurring organization is what I call *structural fractality*: the comparison, its preserved norm, and its projection reappear across the finite hierarchy. The circle is both visible in a lower-dimensional description and retained as a rotational generator within the larger construction.

The next doubling gives sedenions, which have zero divisors. Two nonzero factors can multiply to zero, so the same norm-preserving Hopf construction fails. This identifies a specific limit to the extension: the relationship preserved through the preceding levels no longer survives that multiplication rule (Baez, 2002).

3 What do we measure/observe?

We have described the object and its operations. How do we know the description is right? Writing a claim, understanding it, and showing why it holds are connected activities. A proof

lets someone follow a consequence; a measurement lets them check the relationship from which the account begins.

This is also a question about the foundations of mathematics. What entitles us to trust our starting points? Hilbert sought explicit foundations and a proof of their consistency. Gödel established limits on what sufficiently expressive formal systems can prove from within their own rules. Tarski distinguished the language being examined from the language used to discuss its truth (Gödel, 1931; Tarski, 1944; Zach, 2005). The question of what our statements mean accompanies the question of what follows from them.

The circle brings that inquiry to a relationship we can check by hand. Wrap a string around a round cup. Measure the circumference and the diameter, then divide. Repeat with a larger cup. Within the accuracy of the cup and the instruments, the ratio stays the same. Measure in inches instead of centimetres and both lengths change together; their ratio does not.

To measure is to ask a question in a form an interaction can answer. The string compares lengths, and the ratio expresses their relationship. Another person can repeat the comparison. The exact Euclidean circle gives π ; the physical measurement tells us how closely the cup realizes it. Measuring along a curved surface calls for the geometry of that surface, because the distances being compared have changed (Feynman et al., 1963–1965, Vol. II, Chapter 42).

3.1 Perpendicularity and symmetry

Return to the pizza cut. Once the first diameter is chosen, its perpendicular divides the circle into four right-angle sectors. We can imagine them before marking them. The marks make the relationship visible to someone else; they do not create the right angle.

Hold a horizontal reading fixed and move vertically. The horizontal reading remains unchanged. Reverse the roles and the same is true. Perpendicularity gives this distinction between component changes its geometric form. The square’s diagonal then combines the two: equal sides contribute equally, and its diagonal-to-side ratio remains $\sqrt{2}$ however large or small the square becomes.

Now turn the whole construction. The positions change while the lengths and right angles remain. This is the sameness supplied by symmetry. It concerns a relationship that survives the movement, which is why recognizing the circle does not require its points to stay still.

What value must a quarter-turn have when we write it as an operation? Call the positive quarter-turn i . Two such turns reverse the starting direction, so $i^2 = -1$. Four restore it. The value follows from the turn we specified. The symbol names that operation and lets us compose it with others.

The distinction between the circle and a mark on it also matters. Every rotation preserves the circular boundary as a set. A particular mark returns after a full turn. We can recognize the same circle throughout and still distinguish where the mark has moved.

3.2 Euler’s relationship as a physical law

Why treat this measured relationship differently from the ones we call physical laws? Identify the quantities, perform the operation, and compare the result with its description. The circle lets us do that with perpendicularity, length, angle, and return.

The requirement of consistency is direct: descriptions of the same physical turn must agree about its result. One person writes 90° , another $\pi/2$, another multiplies by i . Once the plane and centre agree, these describe the same turn. The notation changes while the operation remains available.

Euler’s formula keeps the components of that turn together. Its rotation preserves inner products, and therefore lengths, angles, and perpendicularity. At a half-turn it reaches the

opposite pole; at a full turn it restores the marked direction. I call this physically realized relationship of comparison and closure *Euler's law*.

Its spatial and temporal readings belong to the same process. The ratio π relates circumference to diameter. The exponential describes accumulating rotations. A physical angular speed connects angle to elapsed time: at constant speed, $\theta = \omega t$. A continuous traversal then gives a recognizable completion that can be counted. The relationship joins spatial comparison, temporal evolution, and a discrete record of return.

This is already familiar in physics. Oscillations are described by amplitude and phase; electrical phasors keep relative timing in the calculation; coherent contributions at opposite phase cancel. Euler's formula expresses the rotations used in these comparisons (Feynman et al., 1963–1965, Vol. I, Chapter 23; Vol. II, Chapter 22; Vol. III, Chapter 3). Here that common operation becomes the unit through which identity is described and checked.

The circle is minimal for this continuous rotational comparison. A line allows a length-preserving linear sign reversal but no continuous rotation through intermediate directions. A plane does. Following a reference of fixed length through those directions gives the circle. Its simple shape holds the distinction and the preservation needed for the comparison together.

4 How do we convey it?

If I say A, what must you recover? The lighter example already gives an answer. I can say *igniter* and you can say *lighter* while referring to the same thing. I can also repeat your word and point to the wrong object. Matching symbols alone do not establish that we understood one another.

The circle gives us a small case where the agreement can be checked exactly. Draw it, describe a quarter-turn, write 90° , or multiply a complex coordinate by i . These descriptions specify the same operation when their reference and units agree. A receiver who interprets 90 as radians can receive every character and perform the wrong turn. The transmission succeeded; the intended comparison did not.

The representation must let the reader recover the operation it describes. Performing the turn and then describing its result must agree with following the description to calculate the result. This is how the identity remains available through different expressions. The geometric bit makes that requirement explicit in the unit being conveyed.

4.1 What counts as one?

Before counting three cups, what lets each count as one? They may differ in size and colour while remaining cups for the purpose of the count. We distinguish the instances and retain the criterion that makes them instances of the same kind. Number uses that repeatable unit, and changing its notation preserves the quantity counted.

The circle gives a direct example. One full turn restores the starting direction, and a counter records one completed traversal. Turn again: the endpoint is the same and the count is different. Phase tells us where we are within the cycle; the accumulated count tells us how many cycles have occurred. The same picture can carry both when we keep the successive traversals visible.

How much of this do we keep when we make a discrete record? We can name quadrants instead of exact angles. We can name colours within a spectrum or hear separate words in flowing speech. Each boundary retains some differences and treats other variations as the same. Its usefulness depends on what the reader needs to recognize.

Shannon's measure enters at that selection. Two equally likely alternatives give one bit; N equally likely alternatives give $\log_2 N$ bits. With probabilities p_j , entropy is $H =$

$-\sum_j p_j \log_2 p_j$ (Shannon, 1948). A geometric reading supplies distinguishable alternatives, while the preparation supplies their probabilities. The counting problem therefore uses the repeatable identity developed in Section 1: the alternatives must be identifiable before uncertainty among them can be measured.

Closure makes a repeatable identity available; entropy measures uncertainty among the alternatives being distinguished. A reversible change of labels preserves that entropy. Grouping different states into one reading can discard distinctions. The geometric account keeps visible which relationship the record was intended to carry.

If the reconstruction is approximate, we can measure its error. For directions on a circle, the shortest angular separation tells us how far the readings disagree. More finely divided angles require a larger record, and noise can move a received reading across a boundary. The shared unit matters throughout: precision cannot repair an unrecognized difference between degrees and radians.

4.2 Meaning in language and behaviour

How do we tell whether a person understood a relation? We can ask what they can do with it. If one object is larger than another, can they recognize the reverse smaller-than relation? If a second comparison is supplied, can they combine the two? If the context changes, do they follow the appropriate relationship?

Relational Frame Theory studies these questions through mutual entailment, combinatorial entailment, and transformation of stimulus functions (Hayes et al., 2001; Association for Contextual Behavioral Science, n.d.). The first concerns the relation derived in the reverse direction. The second concerns relations derived by combining others. The third concerns how what a stimulus does changes through its learned relations.

For example, learning that a new sign means the same as a warning can give that sign a warning function. Its appearance differs, while its relation to a possible danger guides the response. For larger-than, reversing the comparison gives smaller-than. Understanding includes recognizing what stays the same through the appropriate change of expression or direction.

I ground this account of verbal behaviour in the identity principle: the learner distinguishes expressions and retains the relationship that makes their use coherent. The circle provides an exact case of reversible comparison. Language makes the same question available through learned relations and their effects, which can be checked in what the person does.

Information, meaning, and identity meet here. Something is distinguished; its relationship becomes recognizable; a description makes that relationship available to another person. The geometric bit gives those connected requirements one form. We began with the familiar binary choice, followed its geometry, and made the comparison behind it explicit. We can now return to the lighter and ask what a successful message must preserve: the identity the other person is meant to recover.

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